MILITARY SPECIFICATION

WARHEAD, 2.75 INCH ROCKET, HE, DUAL PURPOSE, M247, PARTS FOR

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements and provides quality assurance provisions for the fabrication of parts and the assembly of one type of warhead designated as Warhead, 2.75 Inch Rocket, HE: M247, Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-48048	Flexible Cable Assembly for M247 Heat Warhead, Assembly and Packing

FSC: 1340

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

FEDERAL

FED-STD-151	Metals, Test Methods

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167	Ammunition Data Cards
MIL-STD-1169	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

A-395-Ferrite Ductile Iron Pressure Retaining Castings for Use at Elevated Temperatures, Specification for.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania).

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9258191	Warhead, 2.75 Inch Rocket, HE: Dual Purpose M247, Metal Parts Assembly

(Copies of specifications, standards, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer.)

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and Assemblies. The components and assemblies shall comply with all requirements secified on Dwg 9258191 and associated drawings, and with all requirements specified in applicable specifications and standards.

3.2.1 Flexible cable assembly. The cable assembly shall comply with specification MIL-F-48048.

3.3 Metal Defects. All components prior to painting or chemical treatment shall be free of surface imperfections on the interior and exterior surfaces of the parts that are indications of discontinuities internal to the metal. (Including are: cracks, lube bursts, pipes, laminations, segregations, porosity, flakes, inclusions and other internal defects.)

3.4 Microexamination. For ductile iron, the microstructure test described in ASTM-A-395 shall be performed for each ladle.

3.5 Continuity and Open Circuit. Each warhead subassembly and each warhead complete assembly shall be subjected to and shall pass the electrical continuity test (less than 0.5 ohms) and open circuit test (greater than 10 megohms).

3.6 Air tightness. The metal parts assembly (after installation of the ogive) shall withstand an air pressure of 10 psig applied to the body cavity for a period of 10 seconds without a pressure drop of more than 0.5 psig.

3.7 Salt Spray Test. The finished assemblies, after coating, shall comply with an accelerated corrosion test.

3.8 First Article Inspection. This specifiction makes provision for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.9 Workmanship. Unless otherwise specified on the drawing, all parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust , porosity, warpage, burn marks, checks, blisters, excess flash, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing base materials. The loading and assembly operations shall be performed in a thorough, workmanlike manner consistent with the best practices of the industry. All required markings shall be neat and sharply defined. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the Contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are made a part of this specification.

4.2 Classification of inspections. Unless otherwise specified herein, or in the contract, the following types of inspection shall be conducted on this item.

a. First Article Inspection
b. Quality Conformance Inspection.

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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PART DESCRIPTION	DRAWING	QUANTITY
Keel Block Samples (If ductile iron is used)		5
Ogive	9271721	15
Cap	9258181	15
Liner	9220009	15
Body	9258180	15
Body Assembly	9260005	20
Warhead, 2.75 Inch Rocket, HE: Dual Purpose M247, Metal Parts Assembly	9258191	25

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50985A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Warhead, 2.75 Inch Rocket, HE: Dual Purpose M247, Metal Parts Assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Keel Block (If Ductile Iron is used) Microexamination	5		3.4	4.5.1
	Ogive (Dwg. 9271721) Examination for defects	15		3.2	4.4.2.1
	Cap (Dwg. 9258181) Examination for defects	15		3.2	4.4.2.2
	Liner (Dwg. 9220009) Examination for defects	15		3.2	4.4.2.3

NOTES:

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TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-W-50985A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Warhead, 2.75 Inch Rocket, HE: Dual Purpose M247, Metal Parts Assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body (Dwg. 9258180)				
	Examination for defects	15		3.2	4.4.2.4
	Warhead casting hardness (Malleable iron)	15		3.1	4.5.2.1
	Warhead casting tension (malleable iron)	6		3.1	4.5.3.1(a)
	Warhead casting hardness (ductile iron)	15		3.1	4.5.2.2
	Warhead casting tension (ductile iron)	6		3.1	4.5.3.2(a)
	Body Assembly (Dwg. 9260005)				
	Examination for defects	20		3.2	4.4.2.5
	Continuity and open circuit	20		3.5	4.4.3.4
	Warhead, 2.75 Inch Rocket, HE: Dual Purpose M247, Metal Parts Assembly (Dwg. 9258191)				
	Examination for defects	25		3.2	4.4.2.6
	Continuity and open circuit	25		3.5	4.5.4
	Air tightness	25		3.6	4.5.5
	Salt spray	25		3.7	4.5.6(b)

NOTES: (a) Three hardest and three softest from hardness test.
(b) If during continuity testing the ogive surface is scratched by the probe, the particular scratched area shall be disregarded.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50985A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9271721
4.4.2.1	Ogive				NEXT HIGHER ASSEMBLY 9259191
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wall thickness, Min		0.40%	3.2	Gage
102	Outside diameter of nose cap end		0.40%	3.2	Gage
103	Contour of ogive nose		0.40%	3.2	Gage
104	Overall length		0.40%	3.2	Gage
105	Inside diameter of base		0.40%	3.2	Gage
106	Inside diameter of hole, forward end		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50985A(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.2	Cap	1	1	9258181
				NEXT HIGHER ASSEMBLY: 9259191

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall thickness, Min		0.40%	3.2	Gage
102	Height of cap forward face		0.40%	3.2	Gage
103	Major diameter threads, min		0.40%	3.2	Gage
104	Pitch diameter threads, max		0.40%	3.2	Gage
105	Diameter of forward face		0.40%	3.2	Gage
Minor					
201	Holes, location		0.65%	3.2	Gage
202	Holes, Inside diameter		0.65%	3.2	Gage
203	Holes, depth		0.65%	3.2	Gage
204	Chamfer		0.65%	3.2	Visual
205	Angle of forward face to cap inner surface		0.65%	3.2	Gage
206	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50985A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9220009
4.4.2.3	Liner				NEXT HIGHER ASSEMBLY 9260005
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Perpendicularity of shoulder to center axis		0.40%	3.2	Gage
102	Outside diameter below shoulder		0.40%	3.2	Gage
103	Concentricity of outside diameter below shoulder		0.40%	3.2	Gage
104	Distance to datum point B		0.40%	3.2	Gage
105	Distance to datum point A		0.40%	3.2	Gage
106	Thickness & variation at other than apex or base sections of cone		0.40%	3.2	Gage
107	Thickness of base flange		0.40%	3.2	Gage
108	Straightness of cone surface		0.40%	3.2	Gage
109	Outside diameter of flange		0.40%	3.2	Gage
110	Surface finish of central portion of cone		0.40%	3.2	Gage
111	Surface finish of outside diameter below flange		0.40%	3.2	Gage
112	Sharp edge present in wire slot		0.40%	3.2	Visual
113	Depth from base of wire slot		0.40%	3.2	Gage
114	Depth from flange of wire slot		0.40	3.2	Gage
115	Thickness of cone at base		0.40%	3.2	Gage
116	Component split, scarred, pitted or dented		0.40%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50985A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.3	Liner				9220009
					NEXT HIGHER ASSEMBLY
					9260005
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
117	Total length of liner		0.40%	3.2	Gage
118	Width of cable slot		0.40	3.2	Gage
Minor					
201	Thickness of cone at apex		0.65%	3.2	Gage
202	Radii improper		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50985A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.2.4	Body				9258180
					NEXT HIGHER ASSEMBLY 9260005
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Warhead front face perpendicular to outside diameter of warhead body		0.40%	3.2	Gage
102	Smallest inside diameter of warhead (forward section)		0.40%	3.2	Gage
103	Runout of inside diameter of warhead (forward section)		0.40%	3.2	Gage
104	Outside diameter of warhead		0.40%	3.2	Gage
105	Outside diameter of forward section		0.40%	3.2	Gage
106	Runout of forward section outside diameter to outside diameter of warhead		0.40%	3.2	Gage
107	True position of base with pitch diameter of base thread		0.40%	3.2	Gage
108	Runout of base with outside diameter of warhead		0.40%	3.2	Gage
109	Runout of bourrelet with outside diameter of warhead		0.40%	3.2	Gage
110	Motor body seat perpendicular to outside diameter of warhead		0.40%	3.2	Gage
111	Pitch diameter of female thread		0.40%	3.2	Gage
112	True position of female thread to outside diameter of warhead		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 9258180
4.4.2.4	Body (Continued)				NEXT HIGHER ASSEMBLY 9260005
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
113	Outside diameter of base		0.40%	3.2	Gage
114	Bourrelet outside diameter		0.40%	3.2	Gage
115	Length of full form threads, min.		0.40%	3.2	Gage
116	Thread damaged or not full		0.40%	3.2	Visual
117	Pitch diameter of acme thread		0.40%	3.2	Gage
118	Major diameter of acme thread, min		0.40%	3.2	Gage
119	Thickness of wall between bourrelet and taper taken at approx. 5 and 8 inches from forward surface and 90° apart (4 measurements)		0.40%	3.2	Gage
120	Runout of aft inside diameter to outside diameter of warhead		0.40%	3.2	Gage
121	Aft inside diameter		0.40%	3.2	Gage
122	Distance from rear face to shoulder forward of internal threads		0.40%	3.2	Gage
123	Taper-forward inside diameter		0.40%	3.2	Gage
124	Surface finish-forward inside diameter		0.40%	3.2	Gage
125	Surface finish-forward outside diameter		0.40%	3.2	Gage
126	Metal defective		0.40%	3.3	Visual
127	Microexamination (Ductile Iron)	(a)		3.4	4.5.1
128	Warhead casting hardness test (Ductile Iron)	(b)		3.1	4.5.2.1

NOTES:

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Body (Continued)	3 OF 3	9258180
			NEXT HIGHER ASSEMBLY 9260005

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
129	Warhead casting Tension (Ductile Iron)	(c)		3.1	4.5.3.1
130	Warhead casting hardness test (Malleable Iron)	(b)		3.1	4.5.2.2
131	Tension Test on Warhead (malleable Iron)	(c)		3.1	4.5.3.2
Minor					
201	Overall length		0.65%	3.2	Gage
202	Length from rear face to rear internal taper		0.65%	3.2	Gage
203	Outside diameter forward of rear external thread		0.65%	3.2	Gage
204	Improper radii or chamfer		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
(a) See 4.5.1
(b) See 4.5.2
(c) See 4.5.3

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PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.5	Body Assembly	1	1	9260005
				NEXT HIGHER ASSEMBLY: 9258191

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Concentricity of liner to warhead body		0.40%	3.2	Gage
102	Liner not fimly attached to warhead body		0.40%	3.2	Visual/Manual
103	Flexible cable assembly not firmly attached to inside wall of warhead body		0.40%	3.2	Visual
104	Flexible cable assembly slot not sealed		0.40%	3.2	Visual
105	Evidence of excess adhesive		0.40%	3.2	Visual
106	Continuity and open circuit		100%	3.5	4.5.4
Minor					
201	Length of flexible cable from aft end		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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MIL-W-50985A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9258191
4.4.2.6	Warhead, 2.75 Inch Rocket, HE: Dual Purpose M247, Metal Parts Assembly				NEXT HIGHER ASSEMBLY -------
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Crimp inadequate		0.40%	3.2	Visual/Manual
102	Ogive loose (torque)		0.40%	3.2	Manual
103	Continuity and open cirucit		100%	3.5	4.5.4
104	Air tightness		0.40%	3.6	4.5.5
105	Salt spray test	(a)		3.7	4.5.6(b)
Minor					
201	Warhead incorrectly or illegibly marked		0.65%	3.2	Visual
202	Finish inadequate or incomplete		0.65%	3.2	Visual
203	Silicone compound missing		0.65%	3.2	Visual
204	Evidence of adhesive on inside of liner		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: (a) Special sampling plan - see 4.5.6
(b) If during continuity testing the ogive surface is scratched by the probe, the particular scratched area shall be disregarded.

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4.4.3 Testing. Testing is described in the Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein. The contractor shall submit all inspection equipment designs to the government for evaluation and approval prior to fabrication of such equipment.

4.5 Test methods and procedures.

4.5.1 Microexamination. For ductile iron, a test lug shall be cast representing the last metal poured from each ladle treated with modularizing agent. The test casting and microexamination shall be in accordance with ASTM-A-395. Failure of the test lug to conform to the microstructure requirements of ASTM-A-395 shall be cause for rejection of the castings from the ladle represented by the test lug.

4.5.2 Hardness test. After final heat treatment, the hardness test shall be performed on the body. Procedures for this test are as prescribed by Federal Test Method Standard No. 151, Method 242.1.

4.5.2.1 Warhead (malleable iron). To insure compliance with the mechanical properties requirements, each body casting from a sample shall be subjected to a hardness test with a Brinell Hardness Tester. The sample, 10 percent minimum, representative of extreme furnace environment, shall be selected from each heat treat batch when a batch type furnace is used for final heat treatment. When a continuous type furnace is used for final heat treatment, from which a 10% minimum sample will also be selected, a heat treat batch shall be considered quantity of castings not to exceed 4 hours production. The castings which constitute the sample shall be collected at regular continuous intervals throughout the 4 hour cycle. The heat treat batch thus constituted shall form the basis for selecting castings for the tension test herein prescribed.

4.5.2.2 Warhead (ductile iron). To insure compliance with the mechanical properties requirements listed in the specifications listed on the warhead drawing, each warhead from a sample shall be subjected to a hardness test with a Brinell Hardness Tester. The sample shall be 1.5% minimum, representative

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of extreme furnace conditions. When a continuous type furnace is used for final heat treatment, a sample shall be 1.5% of the castings completing heat treatment. If boxes or baskets are used which contain approximately 70 castings, one casting from each box shall be tested. A heat treat batch will be considered a quantity of castings not to exceed 8 hours production. If any specimen is outside of the hardness limits, the lot shall be rejected. The lot may be re-heat treated to attain specification requirements.

4.5.3 Tension Test. Two tension test specimens or specimen conforming to a standard "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken longitudinally from each sample. Samples shall be taken 1800 apart. The criteria for determining yield strength and elongation shall be as prescribed by Federal Test Method Standard No. 151, Method 211.1.

4.5.3.1 Warhead (malleable iron). The hardest and softest casting representing the day's heat treat production shall be subjected to this test. Failure of any test specimen to comply with the applicable physical properties requirements on referenced specification of dwg 9258180 shall be cause for rejection of the day's heat treat production represented by the specimens.

4.5.3.2 Warhead (ductile iron). To qualify the heat treatment, one type keel block representative of each heat treat lot shall be subjected to this test. In the case of the continuous heat treat oven, one specimen shall be tested for each (8) hours of oven operating time. If any specimen fails the specification requirements listed on the body drawing, the lot represented by the sample shall be rejected.

4.5.4 Continuity and open circuit test. This test shall be performed 100% on all subassemblies and 100% on all complete metal parts warhead assemblies. Continuity of each conductor circuit shall be checked with an ohmmeter. Insulation resistance between the two fuze attachment contacts shall be checked with a megohm meter. Any assembly that fails to comply with the requirement shall be classed defective and removed from the lot.

4.5.5 Air tightness test. Each warhead assembly in the sample shall be subjected to ten (10) PSIG pressure internally for a period of ten (10) seconds. Observations shall be made by a dry test method with equipment capable of detecting a pressure drop of 0.5 psig. Any warhead assembly that indicates a pressure drop in excess of the requirement shall be classed defective.

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4.5.6 Salt spray test. An accelerated corrosion test shall be performed utilizing the equipment and procedures prescribed by the Federal Test Method Standard No. 151, Method 811. Either a twenty (20) percent concentration of sodium chloride solution shall be used and the length of test shall be one hundred forty four (144) hours or a five (5) percent solution of sodium chloride solution shall be used and the length of test shall be forty-eight (48) hours. Test panels shall not be substituted for the warhead assemblies. The test specimen shall be air dried for a minimum of 96 hours before testing. The surface coated speciment shall be scored through using a sharp knife. The specimens shall then be exposed to the 20 percent salt spray as specified in Federal Test Method Standard No. 151, Method 811.

During production, one completed warhead from each day's production shall be subjected to the Salt Spray Test, until five consecutive day's production have successfully passed the test. Sampling shall then be reduced to one sample once per week. If a failure occurs, the contractor shall be notified to take corrective action. If another consecutive failure occurs after the contractor has been notified to take corrective action, inspection for acceptance shall be stopped until necessary corrective action has been taken to the satisfaction of the Government. After two consecutive failures and after corrective action is taken, sampling shall revert to one specimen from each day's production until five consecutive day's production have again successfully passed the test. Any specimen (warhead) which does not comply with any of the following test criteria shall be considered a failure. The specimen, when subjected to the Salt Spray Test shall show no more than 1/8 inch creepage, blistering, or loss of adhesion of the paint as measured from the scribe mark. At all other points, there shall be no more than a trace of film failure such as blistering or corrosion, and not more than twenty scattered blisters, none larger than 3/16 inch in diameter on the painted surface of the warhead. It is intended that the maximum amount of corrosion in the Salt Spray Test be comparable to photograph 8-1 in ASTM Standard of Evaluation Degree of Resistance to rusting obtained with paint on iron and steel surfaces, D610-68. Photograph 8-1 shall also be used to evaluate surface corrosion on the outer ogive surface.

5. PACKAGING

5.1 Packaging and marking. The warhead shall be packaged and marked in accordance with Military Standard MIL-STD-1169.

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5.2 Packing and marking. The packaged warheads shall be packed and marked in accordance with Military Standard MIL-STD-1169 .

5.3 Ammunition data cards. Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Intended use. The assembly and associated components covered by this specification are intended for use on the 2.75 Inch Folding Fin Aerial Rocket System.

6.2 Ordering Data. See MIL-A-48078(AR).

6.3 Submission of inspection, equipment designs for approval. See 6.2 of MIL-A 48078(AR) Submit designs, as required, to Commander, US Army Armament Research and Development Command, Attn: DRDAR-QAF-I, Dover, NJ 07801.

6.4 First article. When a first article is required for inspection and approval, see 3.8 and 4.3.

6.5 Source control and specification control drawings. Certificates of conformance from the supplier certifying that the item complies with the drawing requirements shall be provided to the government representative.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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